4 Industrial wastewater is wastewater produced by industrial facilities during manufacturing or other processes. It differs from domestic wastewater due to the presence of a wider range of pollutants and the potential for higher concentrations of contaminants.

(a) An industrial facility discharges wastewater containing Al^{3+}, Zn^{2+}, and Cu^{2+} ions. To meet environmental standards, the facility designs a treatment process by controlled precipitation to remove these ions from the wastewater.

(i) Table 4.1 shows the K_{sp} values for the hydroxides of the metal ions.
Given that the wastewater initially contains 0.010 mol dm^{-3} of each metal ion, calculate the minimum pH at which copper(II) hydroxide begin to precipitate.

Table 4.1

metal hydroxide	K_{sp}	minimum pH for precipitation
$Al(OH)_3$	1.3×10^{-33}	3.70
$Cu(OH)_2$	2.2×10^{-20}	–
$Zn(OH)_2$	3.0×10^{-17}	6.74

[2]

(ii) Based on your calculations in **(a)(i)**, explain how Al^{3+}, Cu^{2+} and Zn^{2+} ions in the wastewater can be separated in the treatment process.

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(b) Industrial wastewater containing Cu^{2+} ions must be treated before it can be safe for discharge to a sanitary sewer for further treatment. The allowable limit of Cu^{2+} in the discharge is 1 mg per litre.

(i) Calculate the concentration of Cu^{2+} ions in the wastewater when zinc(II) hydroxide starts to precipitate.

[2]

(ii) The industrial wastewater is treated at pH 6.74. Using your answer in **(b)(i)**, deduce if the wastewater is safe for discharge to sanitary sewage for further treatment?

[1]

(c) There may be difficulty in removing Al^{3+} ion from the wastewater when excess sodium hydroxide solution is added.

Explain, with the aid of suitable equation(s), why this is so.

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(d) A sample of the industrial wastewater was found to be pale blue in colour due to contamination by Cu^{2+} ions.

(i) Explain why the solution contaminated with Cu^{2+} is blue in colour.

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(ii) A student took a sample of the solution and added excess concentrated hydrochloric acid. State the type of reaction and describe any observations. Construct a balanced equation for the reaction.

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[total: 14]